Using Propensity Scores

Lucy D'Agostino McGowan

Wake Forest University

Propensity scores

Matching

Weighting

Stratification

Direct Adjustment

...

estimand

Ingredients

- 150g unsalted butter, plus extra for greasing
- 150g plain chocolate, broken into pieces
- 150g plain flour
- ½ tsp baking powder
- ½ tsp bicarbonate of soda
- 200g light muscovado sugar
- 2 large eggs

Method

1. Heat the oven to 160C/140C fan/gas 3. Grease and base line a 1 litre heatproof glass pudding basin and a 450g loaf tin with baking parchment.

2. Put the butter and chocolate into a saucepan and melt over a low heat, stirring. When the chocolate has all melted remove from the heat.

estimator

estimate

Image source: Simon Grund

Propensity scores

Matching

Weighting

Stratification

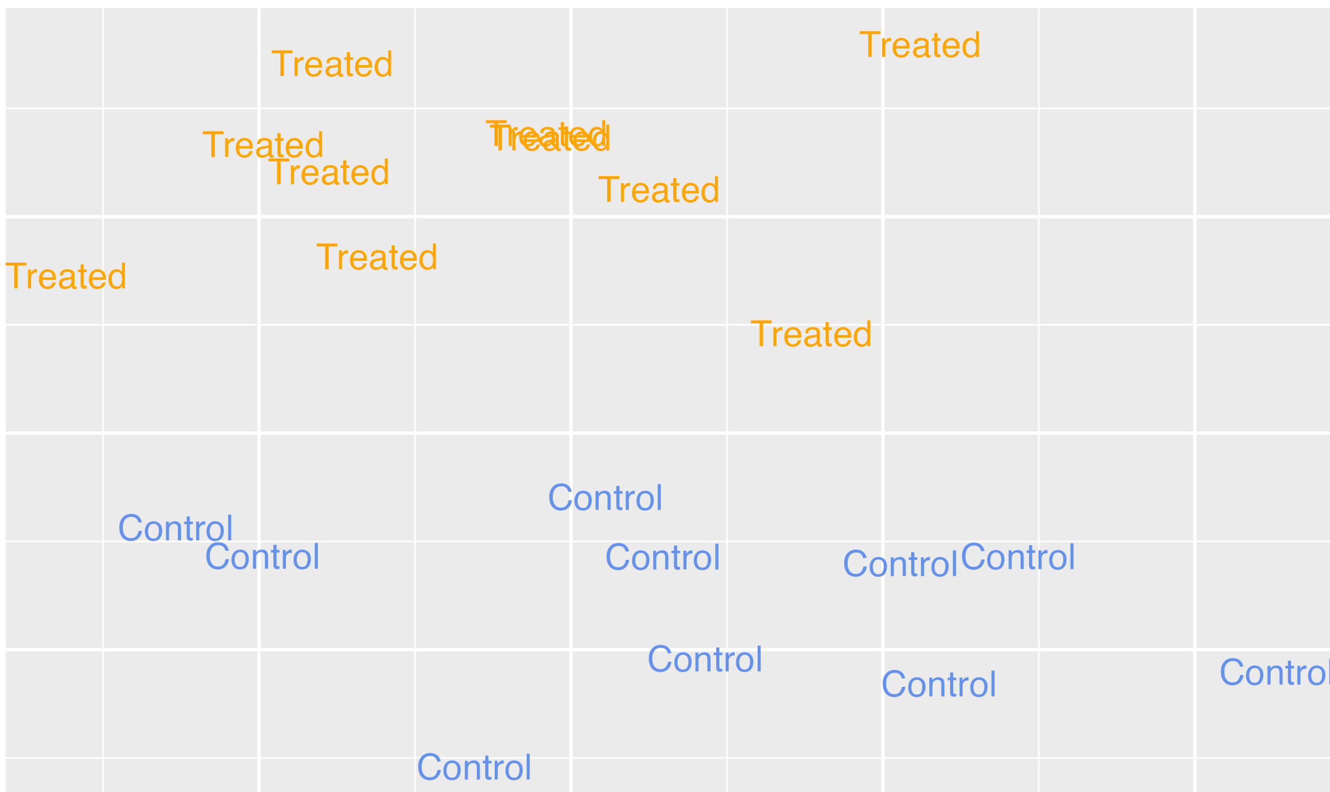
Direct Adjustment

…

Target estimands

Average Treatment Effect (ATE)

$$\tau = E[Y(1) - Y(0)]$$



Target estimands

Estimand	Target population	Example Research Question
ATE	Full population	*Should we decide whether to have extra magic hours all mornings to change the posted wait time for Seven Dwarfs Mine Train between 9-10 AM?* *Should a specific policy be applied to all eligible observations?*

Target estimands

Average Treatment Effect among the Treated (ATT)

$$\tau = E[Y(1) - Y(0)|Z = 1]$$

Target estimands

Estimand	Target population	Example Research Question
ATT	Exposed (treated) observations	*Should we stop extra magic hours to change the posted wait time for Seven Dwarfs Mine Train between 9-10 AM?* *Should we stop our marketing campaign to those currently receiving it?* *Should medical providers stop recommending treatment for those currently receiving it?*

Matching in R (ATT)

```
library(MatchIt)
m <- matchit(
  net_num ~ income + health + temperature,
  data = net_data
)
m
```

```
A `matchit` object
 - method: 1:1 nearest neighbor matching without replacement
 - distance: Propensity score
             - estimated with logistic regression
 - number of obs.: 1752 (original), 908 (matched)
 - target estimand: ATT
 - covariates: income, health, temperature
```

Matching in R (ATT)

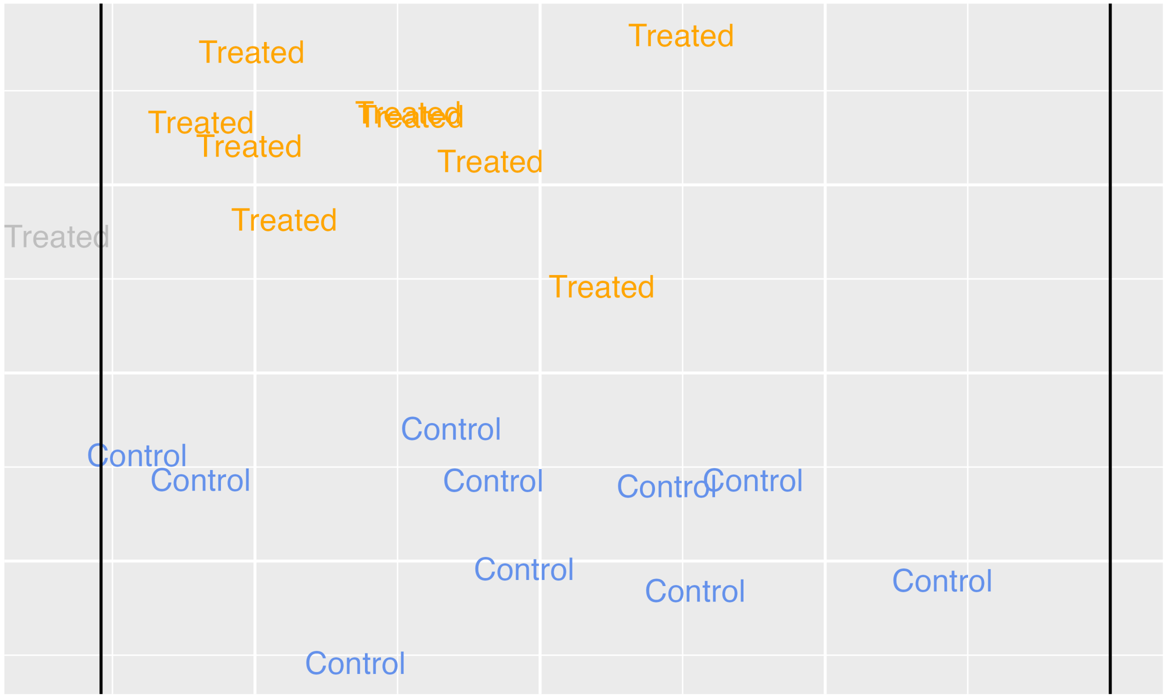
```r
matched_data <- get_matches(m, id = "i")
as_tibble(matched_data)
```

```
# A tibble: 908 × 14
   i     subclass weights    id net   net_num malaria_risk
   <chr> <fct>      <dbl> <int> <lgl>   <int>        <dbl>
 1 16    1              1    16 TRUE        1           15
 2 109   1              1   109 FALSE       0           22
 3 18    2              1    18 TRUE        1           28
 4 1652  2              1  1652 FALSE       0           39
 5 19    3              1    19 TRUE        1           18
 6 64    3              1    64 FALSE       0           22
 7 22    4              1    22 TRUE        1           17
 8 329   4              1   329 FALSE       0           22
 9 28    5              1    28 TRUE        1           22
10 706   5              1   706 FALSE       0           25
```

Target estimands

Average Treatment Effect among the Unexposed (ATU / ATC)

$$\tau = E[Y(1) - Y(0)|Z = 0]$$



Target estimands

Estimand	Target population	Example Research Question
ATU	Unexposed (control) observations	*Should we add extra magic hours for all days to change the posted wait time for Seven Dwarfs Mine Train between 9-10 AM?* *Should we extend our marketing campaign to those not receiving it?* *Should medical providers extend treatment to those not currently receiving it?*

Matching in R (ATU)

```r
m <- matchit(
  net_num ~ income + health + temperature,
  data = net_data,
  estimand = "ATC"
)
m
```

```
A `matchit` object
 - method: 1:1 nearest neighbor matching without replacement
 - distance: Propensity score
             - estimated with logistic regression
 - number of obs.: 1752 (original), 908 (matched)
 - target estimand: ATC
 - covariates: income, health, temperature
```

`matchit()` spells this estimand `ATC`

Target estimands

Average Treatment Effect among the Matched (ATM)

Target estimands

Estimand	Target population	Example Research Question
ATM	Evenly matchable	*Are there some days we should change whether we are offering extra magic hours in order to change the posted wait time for Seven Dwarfs Mine Train between 9-10 AM?* *Is there an effect of the exposure for some observations?* *Should those at clinical equipoise receive treatment?*

Matching in R (ATM)

```r
m <- matchit(
  net_num ~ income + health + temperature,
  data = net_data,
  link = "linear.logit",
  caliper = 0.1
)
m
```

The caliper is the largest difference between two observations we accept as a match, measured in standard deviations of the propensity score (here, on the linear logit scale). Observations that cannot be matched within it are discarded

Matching in R (ATM)

```
A `matchit` object
 - method: 1:1 nearest neighbor matching without replacement
 - distance: Propensity score [caliper]

             - estimated with logistic regression and linearized
 - caliper: <distance> (0.036)
 - number of obs.: 1752 (original), 898 (matched)
 - target estimand: ATT
 - covariates: income, health, temperature
```

Matching in R (ATM)

```r
matched_data <- get_matches(m, id = "i")
as_tibble(matched_data)
```

```
# A tibble: 898 × 14
   i     subclass weights    id net   net_num malaria_risk
   <chr> <fct>      <dbl> <int> <lgl>   <int>        <dbl>
 1 16    1              1    16 TRUE        1           15
 2 109   1              1   109 FALSE       0           22
 3 18    2              1    18 TRUE        1           28
 4 1652  2              1  1652 FALSE       0           39
 5 19    3              1    19 TRUE        1           18
 6 657   3              1   657 FALSE       0           22
 7 22    4              1    22 TRUE        1           17
 8 329   4              1   329 FALSE       0           22
 9 28    5              1    28 TRUE        1           22
10 706   5              1   706 FALSE       0           25
```

Your Turn 1

Using the propensity scores you created in the previous exercise, create a “matched” data set using the ATM method with a caliper of 0.2.

Propensity scores

Matching

Weighting

Stratification

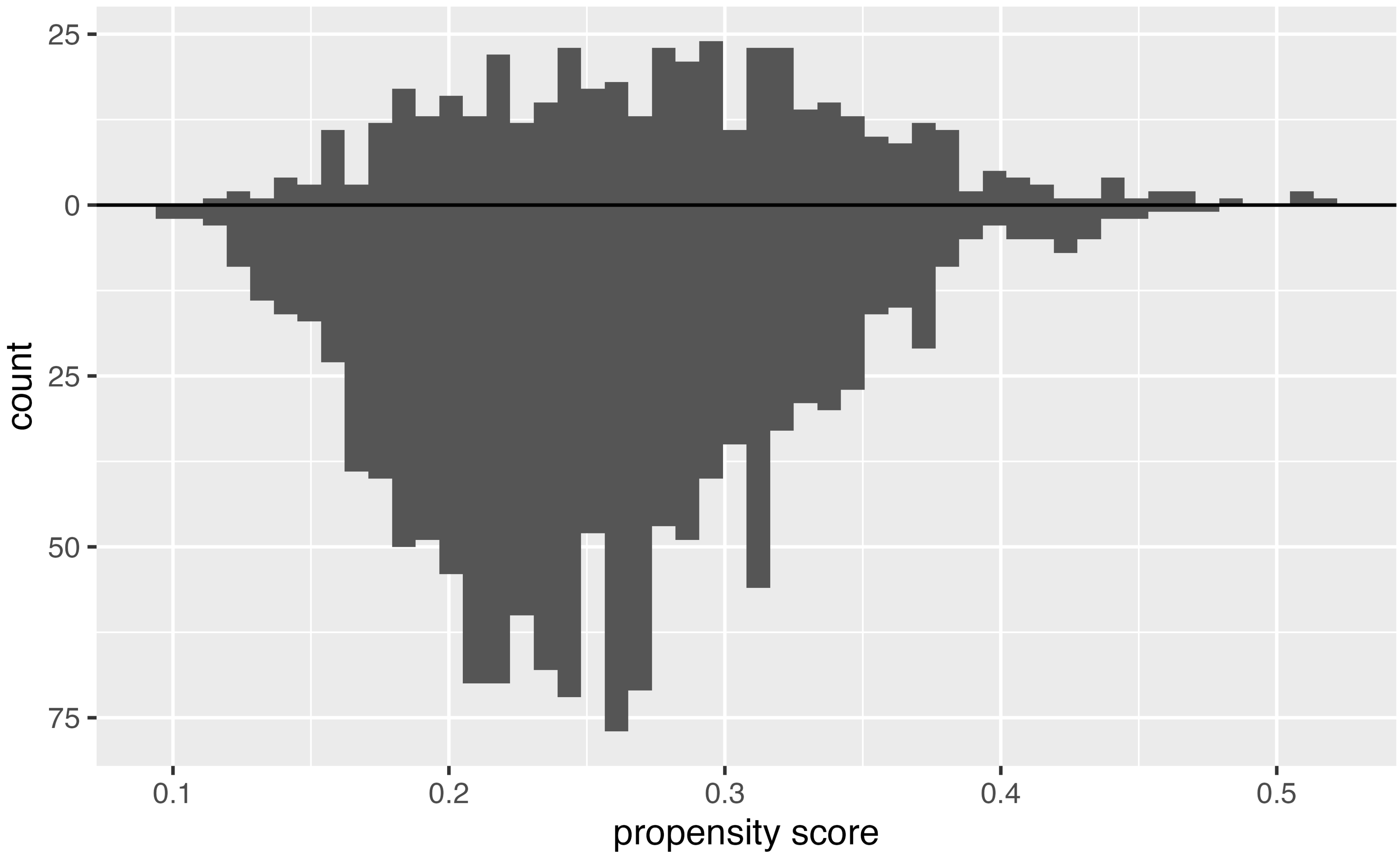
Direct Adjustment

…

Matching or weighting?

- Weighting is statistically more efficient; we recommend it when possible
- Matching has a distinct advantage: it's easy to understand
- Not everyone will follow a pseudo-population or a non-integer sample size

Histogram of propensity scores



Tilting functions

Every weight is the ATE weight times a **tilting function**, $h(p_i)$

$$w_i = \frac{h(p_i)Z_i}{p_i} + \frac{h(p_i)(1 - Z_i)}{1 - p_i}$$

The tilting function reshapes the population we are targeting

Target estimands: ATE

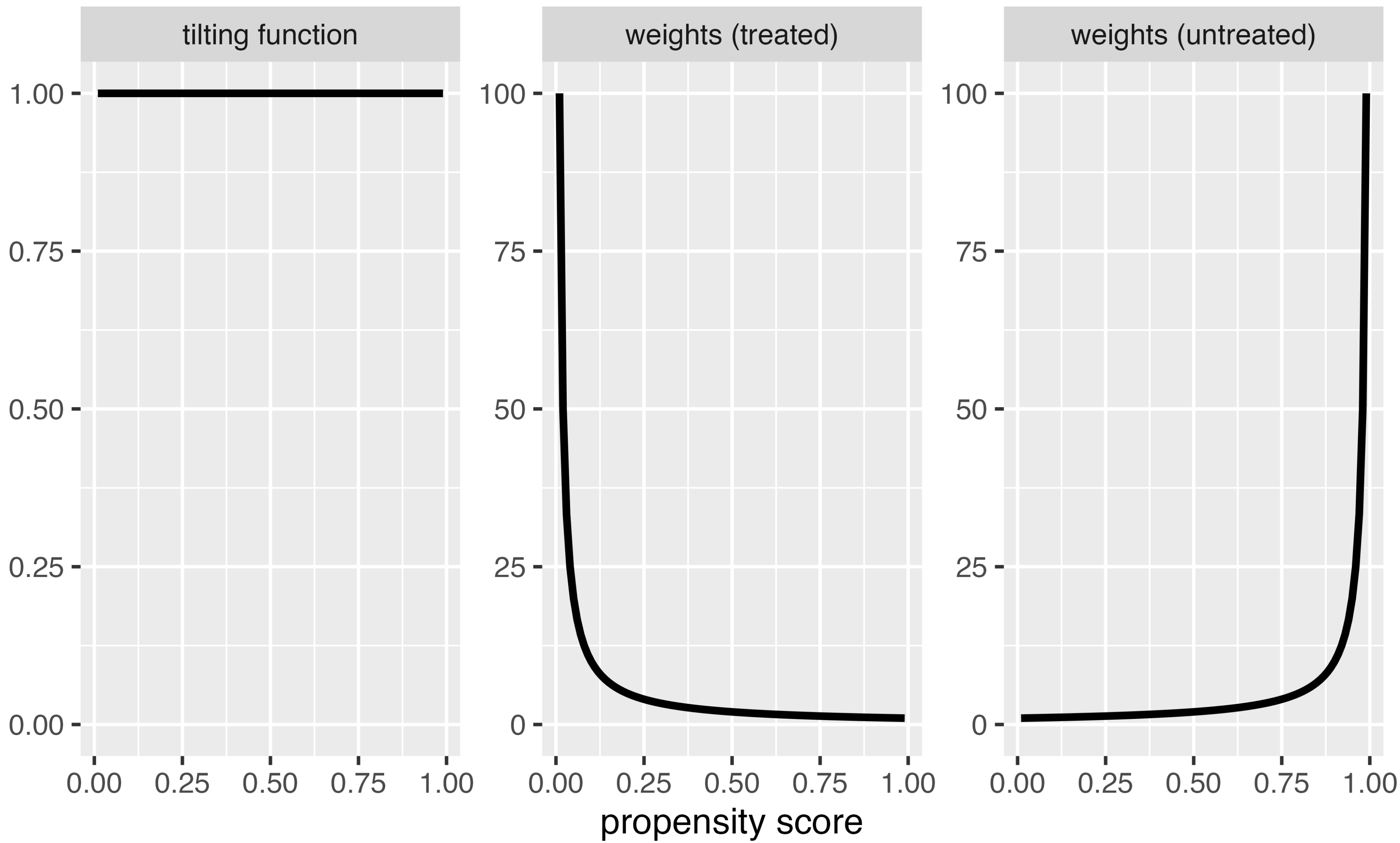
Average Treatment Effect (ATE)

$$w_{ATE} = \frac{Z_i}{p_i} + \frac{1 - Z_i}{1 - p_i}$$

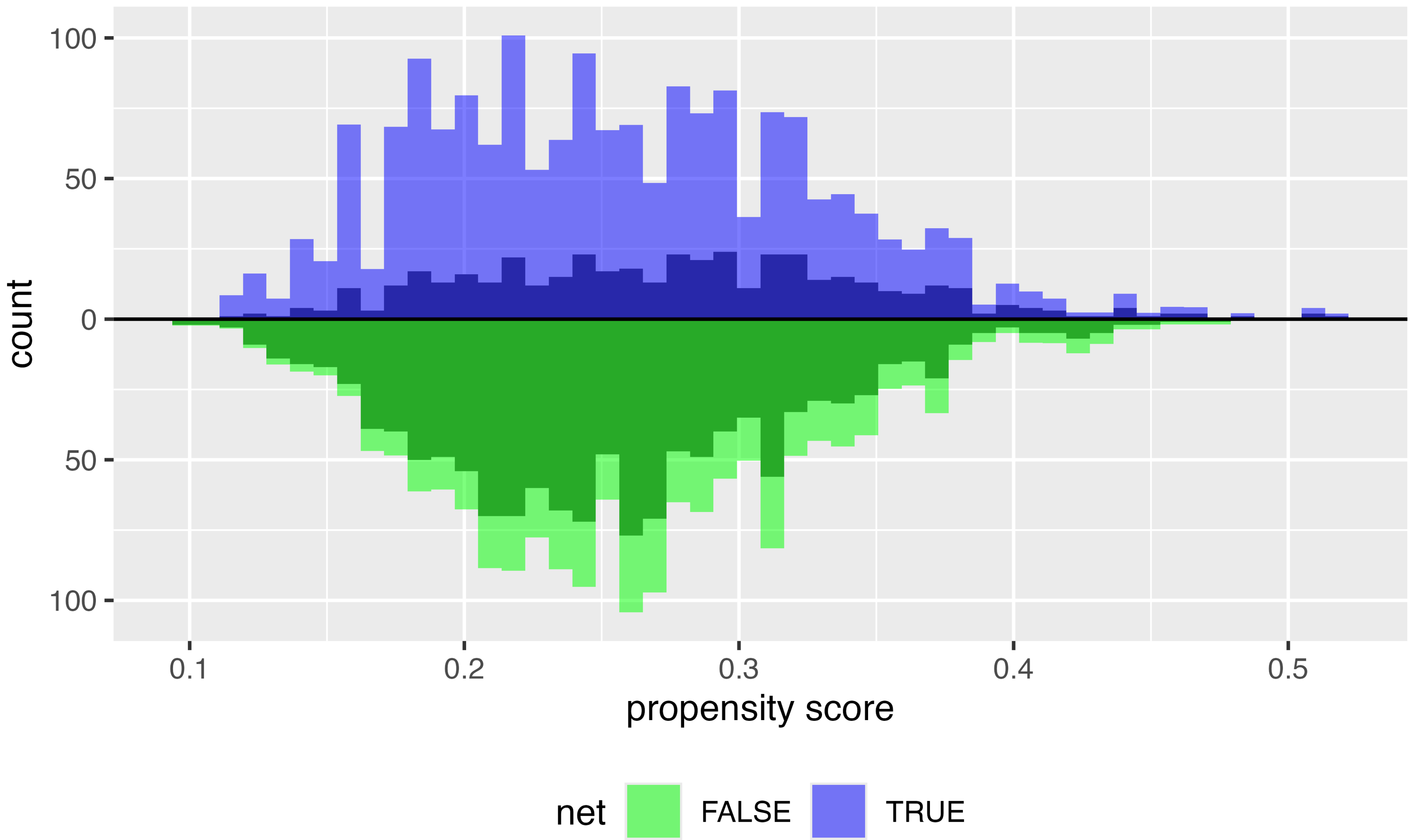
```
(Z / p) + ((1 - Z) / (1 - p))
```

The tilting function is $h(p_i) = 1$: the population is unchanged and the weights are unbounded

ATE tilting function



ATE



100
50
0
50
100
count
0.1
0.2
0.3
0.4
0.5
propensity score
net
FALSE
TRUE

Target estimands: ATT & ATU

Average Treatment Effect Among the Treated (ATT)

$$w_{ATT} = \frac{p_i Z_i}{p_i} + \frac{p_i(1 - Z_i)}{1 - p_i}$$

```
((p * Z) / p) + ((p * (1 - Z)) / (1 - p))
```

The tilting function is $h(p_i) = p_i$: the treated get a weight of 1

Target estimands: ATT & ATU

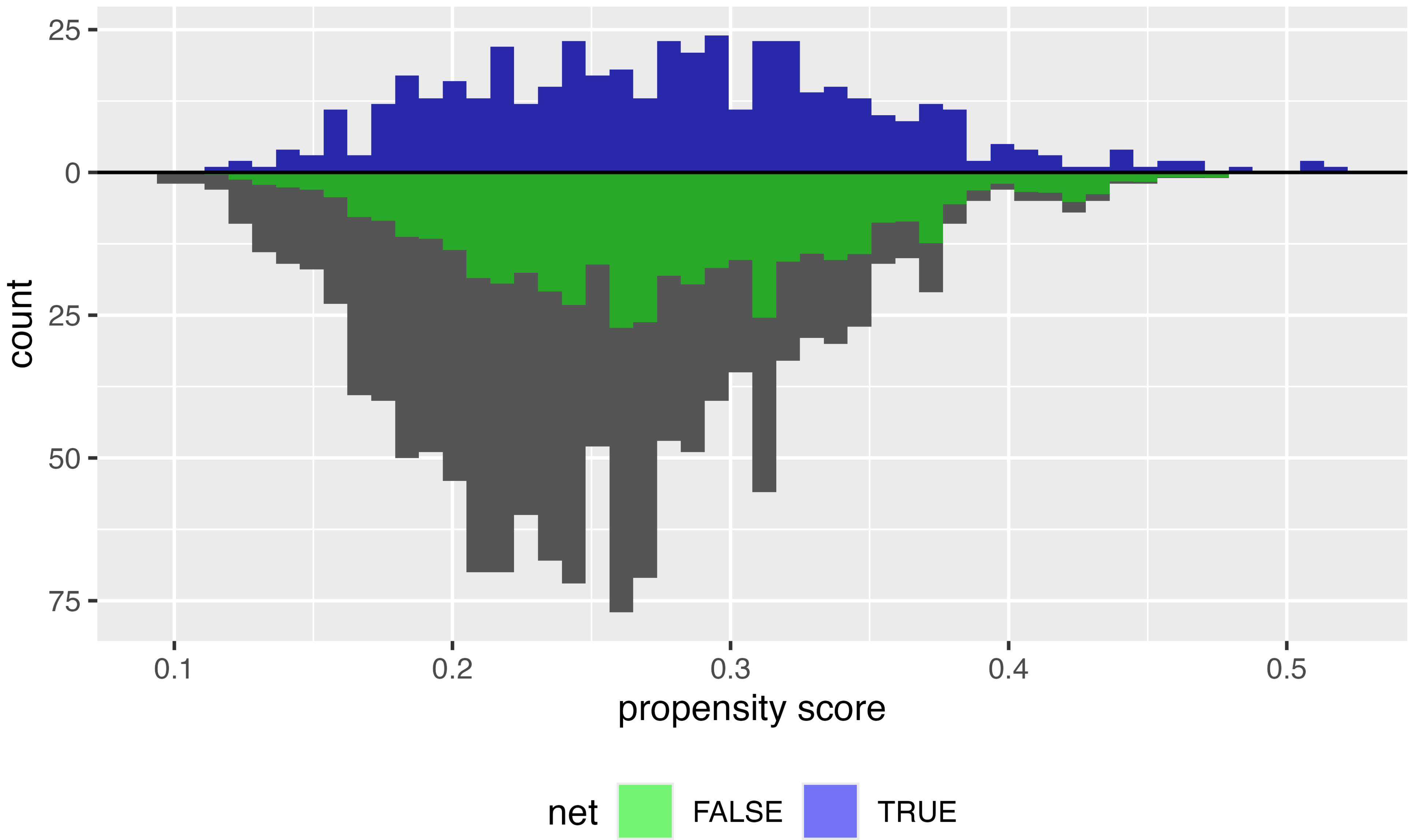
Average Treatment Effect Among the Unexposed (ATU)

$$w_{ATU} = \frac{(1 - p_i)Z_i}{p_i} + \frac{(1 - p_i)(1 - Z_i)}{(1 - p_i)}$$

```
(((1 - p) * Z) / p) + (((1 - p) * (1 - Z)) / (1 - p))
```

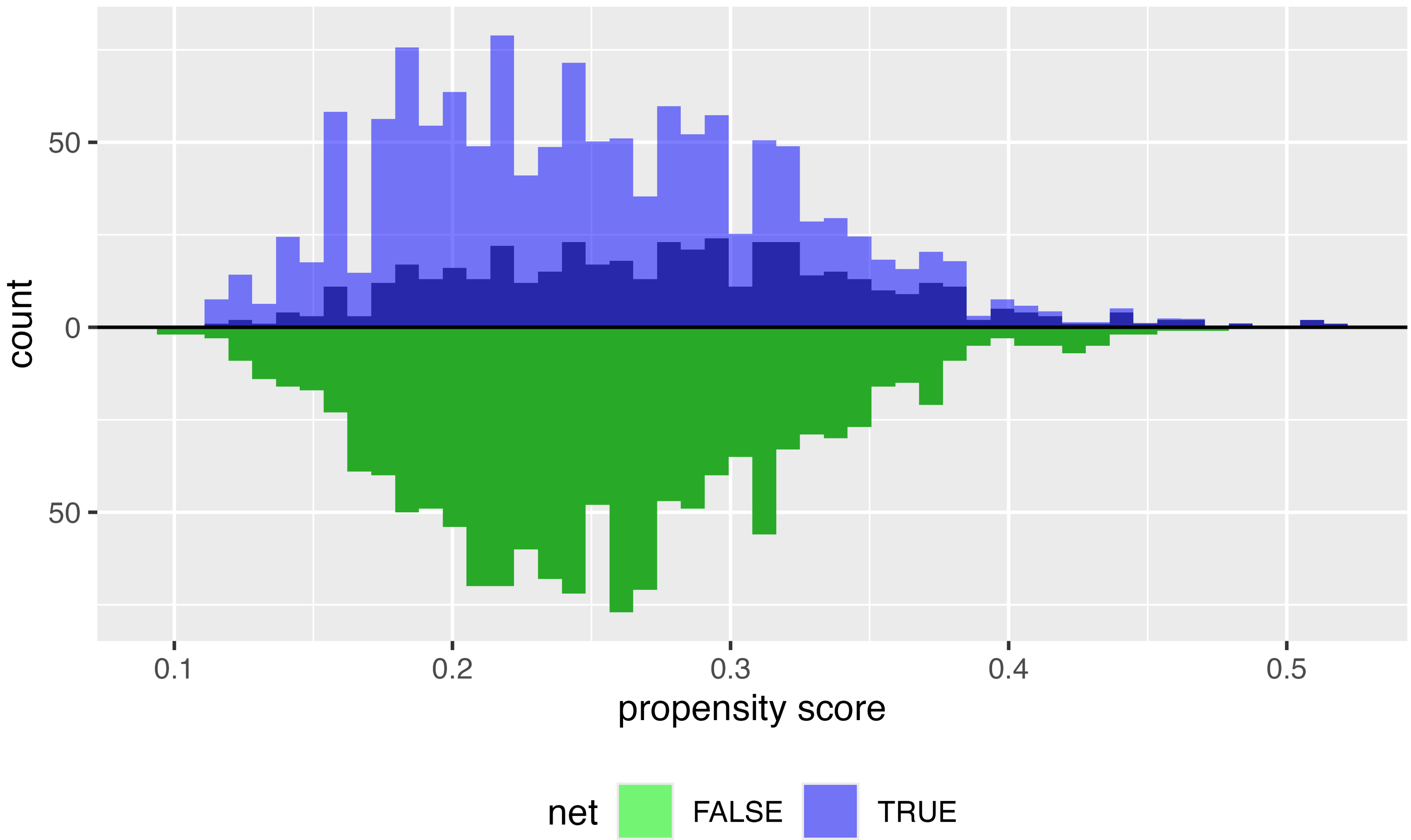
The tilting function is $h(p_i) = 1 - p_i$: the untreated get a weight of 1

ATT



25
0
25
50
75
count
0.1
0.2
0.3
0.4
0.5
propensity score
net
FALSE
TRUE

ATU



Target estimands: ATM & ATO

Average Treatment Effect Among the Evenly Matchable (ATM)

$$w_{ATM} = \frac{\min\{p_i, 1 - p_i\}}{Z_i p_i + (1 - Z_i)(1 - p_i)}$$

```r
pmin(p, 1 - p) / (Z * p + (1 - Z) * (1 - p))
```

The tilting function is $h(p_i) = \min\{p_i, 1 - p_i\}$: a triangle peaking at 0.5

Target estimands: ATM & ATO

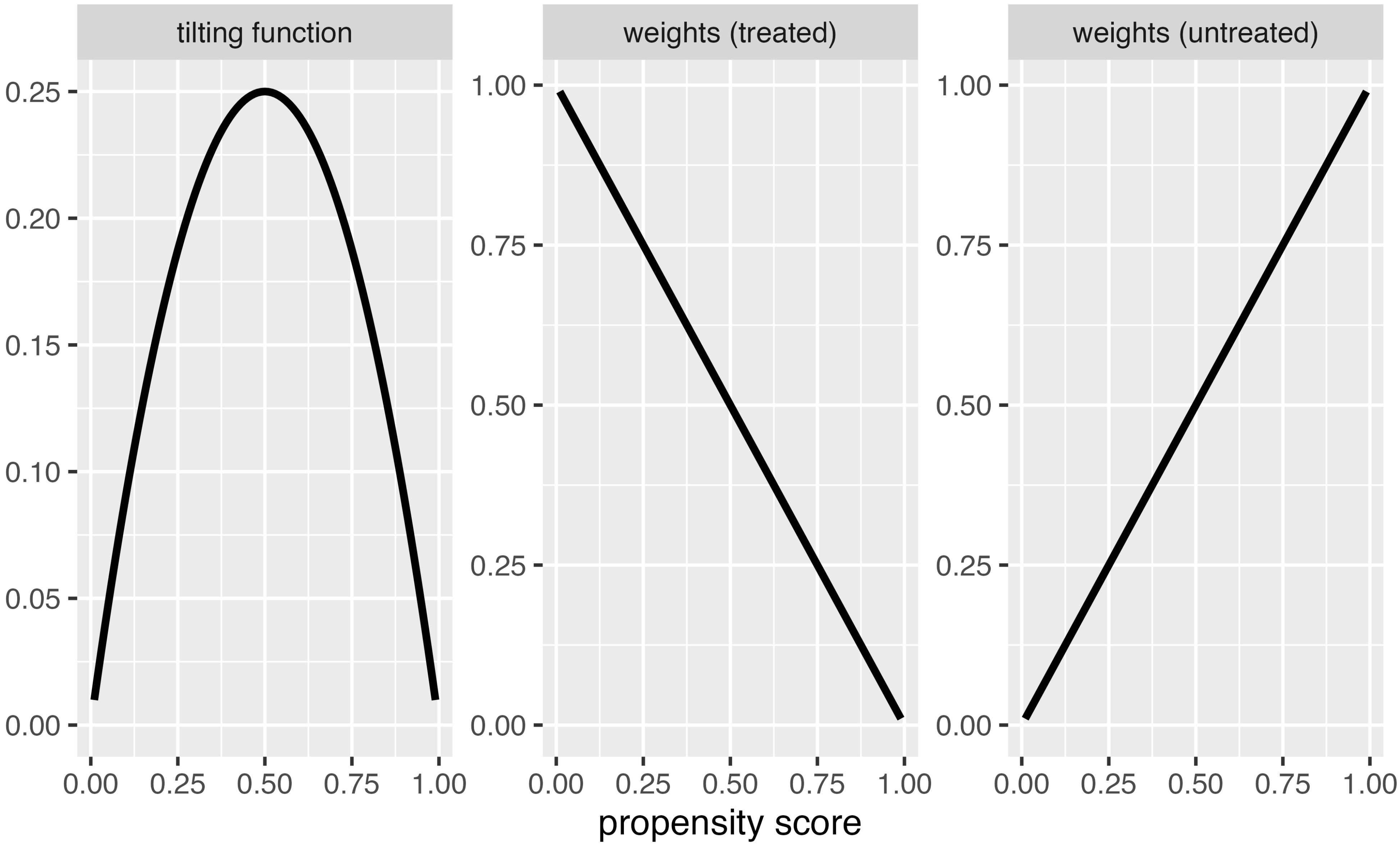
Average Treatment Effect Among the Overlap Population

$$w_{ATO} = (1 - p_i)Z_i + p_i(1 - Z_i)$$

```
(1 - p) * Z + p * (1 - Z)
```

The tilting function is $h(p_i) = p_i(1 - p_i)$: smooth, peaking at 0.5, and the weights are bounded by 0 and 1

ATO tilting function

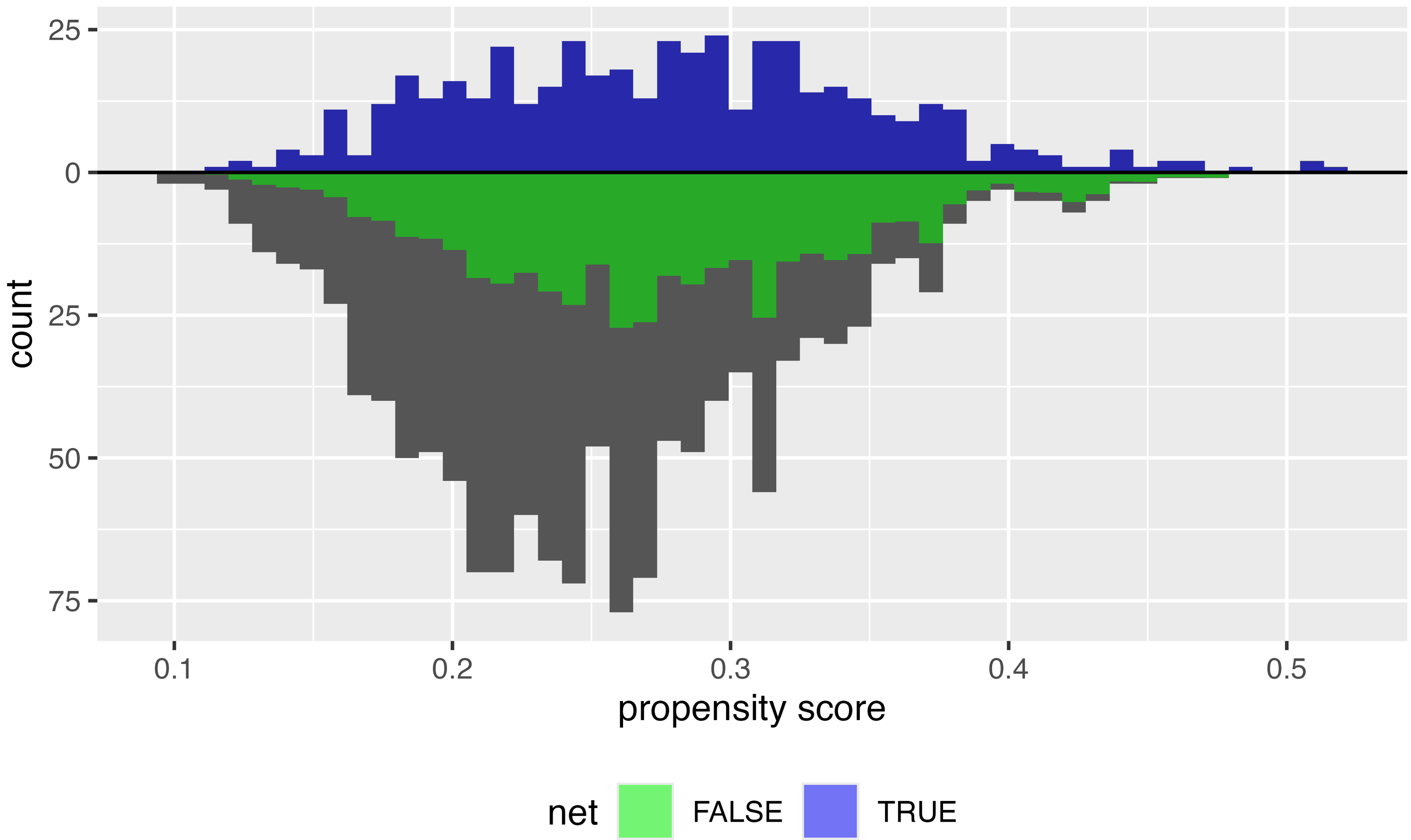


Target estimands

Estimand	Target population	Example Research Question
ATO	Overlap population	*Same as ATM*

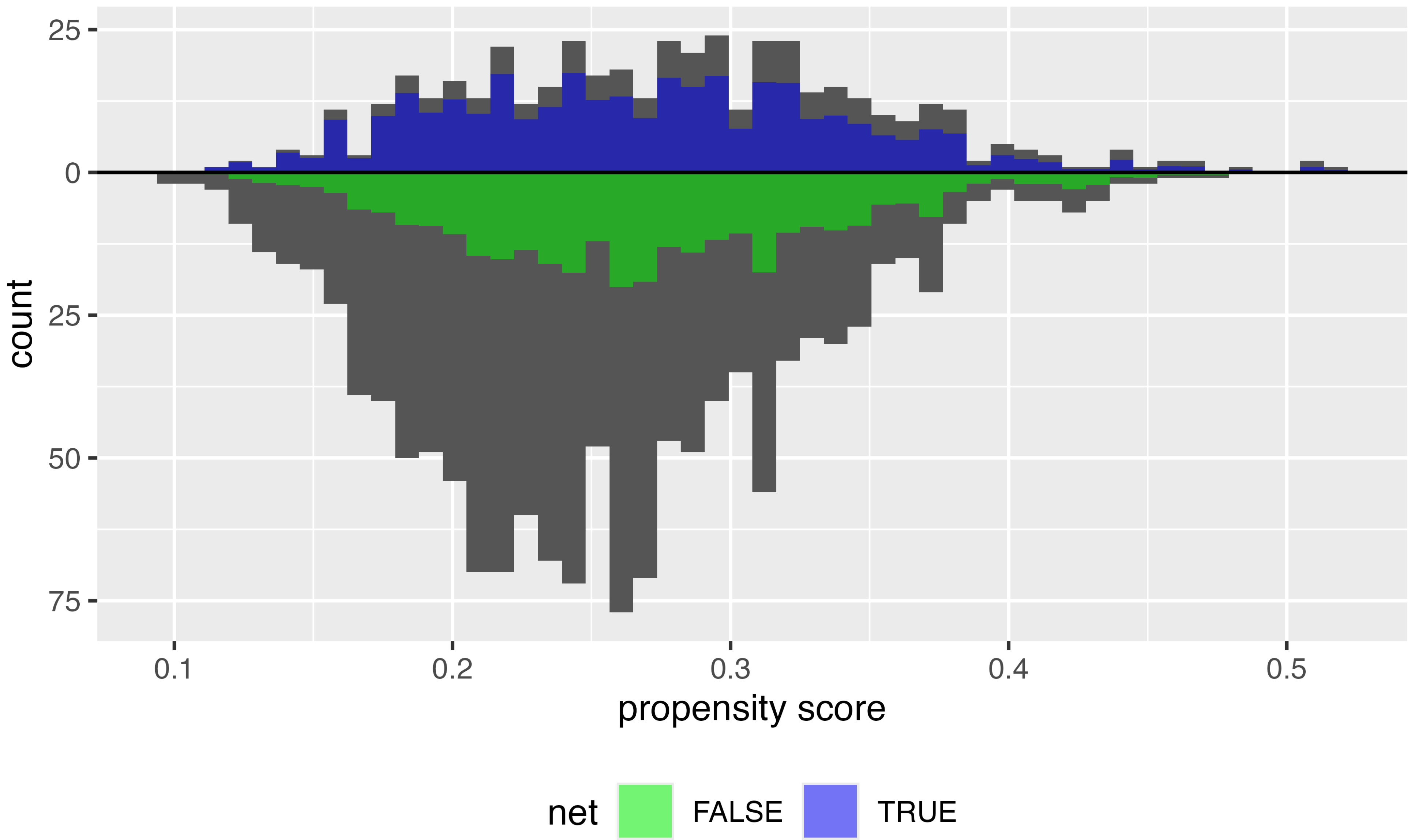
We will compare these populations with the effective sample size when we check balance

ATM



25
0
25
50
75
count
0.1
0.2
0.3
0.4
0.5
propensity score
net
FALSE
TRUE

ATO



25
0
25
50
75
count
0.1
0.2
0.3
0.4
0.5
propensity score
net
FALSE
TRUE

ATE in R

Average Treatment Effect (ATE) $w_{ATE} = \frac{Z_i}{p_i} + \frac{1-Z_i}{1-p_i}$

```r
library(propensity)
df <- propensity_model |>
  augment(type.predict = "response", data = net_data) |>
  mutate(w_ate = wt_ate(.fitted, net))
```

Extreme weights

- The minimum of the ATE weights is 1, but the maximum is unbounded

- The closer the probability of the observed exposure is to 0, the higher the weight

- **Extreme weights** *destabilize* our estimate, resulting in worse precision and potentially worse bias

Extreme weights

```
df |>
  arrange(desc(w_ate)) |>
  select(net, income, health, temperature, .fitted, w_ate)
```

Extreme weights

```
# A tibble: 1,752 × 6
   net   income health temperature .fitted       w_ate
   <lgl>  <dbl>  <dbl>       <dbl>   <dbl> <psw{ate}>
 1 TRUE     501     10        29.9   0.117   8.556008
 2 TRUE     456      7        26.4   0.120   8.328450
 3 TRUE     544     10        29.7   0.126   7.915342
 4 TRUE     424     13        20.2   0.136   7.361471
 5 TRUE     613     43        30.4   0.138   7.270849
 6 TRUE     644     33        31.9   0.139   7.190567
 7 TRUE     532     26        24.6   0.143   7.011099
 8 TRUE     502     21        22.7   0.143   6.968780
 9 TRUE     633     35        29.7   0.145   6.883178
10 TRUE     547     18        24.7   0.146   6.845732
```

Stabilized weights

```r
df <- df |>
  mutate(
    sw_ate = wt_ate(.fitted, net, stabilize = TRUE)
  )

df |>
  summarize(mean = mean(sw_ate), min = min(sw_ate), max = max(sw_ate))
```

```
# A tibble: 1 × 3
   mean   min   max
  <dbl> <dbl> <dbl>
1  1.00 0.501  2.22
```

Stabilizing puts the marginal probability of the observed exposure in the numerator: the proportion exposed for exposed units, the proportion unexposed for unexposed units

Trimming and truncation

- **Trimming**: set an acceptable range of the propensity score and drop the observations outside it
- **Truncation**: set the values outside the range to the minimum or maximum of the range
- Some authors use these terms interchangeably, or with the opposite meanings

Trimming and truncation in R

```
df <- df |>
  mutate(
    trimmed_ps = ps_trim(.fitted, method = "adaptive") |>
      ps_refit(propensity_model),
    trimmed = is_unit_trimmed(trimmed_ps),
    trunc_ps = ps_trunc(.fitted, method = "pctl", lower = 0.01, upper = 1)
  )

df |>
  select(net, income, health, temperature, .fitted, trimmed) |>
  filter(trimmed)
```

Trimming and truncation in R

```
# A tibble: 2 × 6
  net   income health temperature .fitted trimmed
  <lgl>  <dbl>  <dbl>       <dbl>   <dbl> <lgl>
1 FALSE    340     15        26.3  0.0987 TRUE
2 FALSE    351     21        27.1  0.0980 TRUE
```

Trimming and truncation

- Trimming and truncation, like using a caliper, change the population we are making inferences about

- The tilting function is 0 for the observations we trimmed and 1 for the ones that remain

- You can think of the resulting estimand as another type of overlap population

Your Turn 2

Using the propensity scores you created in the previous exercise, add the ATE weights to your data frame

***Stretch*: Using the same propensity scores, create ATM weights**

06:00